

Request Processing Database ERD

PostgreSQL database: hrms_requests

requests		
Key	Column	Type
PK	id	UUID
	user_id	UUID
	request_type	VARCHAR(100)
	payload	TEXT
	status	VARCHAR(30)
	result	TEXT
	error_message	TEXT
	created_at	TIMESTAMPTZ
	updated_at	TIMESTAMPTZ
	completed_at	TIMESTAMPTZ
	version	BIGINT
	idempotency_key	VARCHAR(100)
	idempotency_fingerprint	VARCHAR(64)

Constraints

- UNIQUE (user_id, idempotency_key)
- status ∈ { PENDING, PROCESSING, COMPLETED, FAILED }
- version ≥ 0



Physical FK

processed_events		
Key	Column	Type
PK FK	event_id	UUID
	request_id	UUID
	event_type	VARCHAR(100)
	event_version	INTEGER
	occurred_at	TIMESTAMPTZ
	processed_at	TIMESTAMPTZ

Constraints

- FOREIGN KEY (request_id) REFERENCES requests(id) ON DELETE CASCADE
- event_version > 0



Logical Reference
(No Physical FK)

outbox_events		
Key	Column	Type
PK	id	UUID
	aggregate_id	UUID
	aggregate_type	VARCHAR(100)
	event_type	VARCHAR(100)
	payload	JSONB
	status	VARCHAR(30)
	retry_count	INTEGER
	created_at	TIMESTAMPTZ
	published_at	TIMESTAMPTZ
	last_error	TEXT

Constraints

- status ∈ { PENDING, PROCESSING, PUBLISHED, FAILED }
- retry_count ≥ 0

Legend



= Physical Foreign Key



= Logical Reference (No Physical Foreign Key)

PK = Primary Key

FK = Foreign Key

UQ = Unique Constraint

Notes

- requests.user_id is a logical identity from Auth domain (no physical foreign key).
- outbox_events.aggregate_id logically identifies the request aggregate.
- processed_events records processed event IDs to prevent duplicate event processing.
- All timestamp columns use PostgreSQL TIMESTAMPTZ.